

COMP3018 Probabilistic Machine Learning

Artefact 2 Assessment Rubric

	High Distinction	Distinction	Credit	Pass	Fail
Technical Report (27%)					
Individual Component	Outstanding report. Exceptional justification for non-i.i.d. sequential modelling. Excellent evaluation of Markov models, state transitions, and inference algorithms. Deep, critical analysis of computational complexity, scalability, and system brittleness.	Strong, well-structured report. Clear, accurate rationale for Markov models. Solid evaluation of state/emission probabilities, inference algorithms, and theoretical computational complexity. Effectively analyses system performance.	Well-written report. Adequate integration of sequential Markov approaches. Discusses state spaces and algorithms correctly, but lacks deep critical insight into computational scalability or failure modes.	Satisfactory report. Basic concepts of sequential modelling are present. Minimal justification of algorithms or complexity trade-offs. Stress testing or evaluation of uncertainty is brief or incomplete.	Poorly written or incomplete. Fails to understand sequential/Markov foundations. Missing substantive analysis of algorithms or ignores computational complexity entirely.
Technical Implementation (30%)					
Individual Component	Sophisticated, robust sequential architecture. Excellent implementation of Markov models and inference algorithms (e.g., Viterbi). Code is highly optimised for scalability, handles sequential uncertainty flawlessly, and meets professional standards.	Well-designed implementation. Strong use of probabilistic libraries to model state transitions and perform sequential inference. Code is automated, functional, and follows best practices.	Logical implementation meeting core requirements. Models function and generate correct sequential inferences, but may lack computational efficiency or elegant handling of state space transitions.	Meets minimum requirements. Generates basic sequential outputs but struggles with state management, exhibits poor code structure, or relies on inefficient inference loops.	Architecture is insufficient or incomplete. Code is nonfunctional or fails to implement the required Markov/sequential models.

Student Name:	Overall Mark:	Grade:
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Video Demonstration (5%)					
Individual Component	Highly professional, clear demonstration within the 5-7 minute limit. Excellent technical quality, providing detailed insights into state transitions, uncertainty over time, complexity insights, and stakeholder resolution.	Clear, well-presented demonstration. Adheres to the time limit. Effectively explains the automated pipeline, sequential inference execution, and how uncertainty changes over time.	Demonstrates core functionality but presentation might lack clarity in explaining state transitions or complexity insights. Acceptable technical quality.	Demonstration is difficult to follow, exceeds time limits, or doesn't clearly show the Markov model handling sequential data. Poor technical quality.	Missing, incomplete, or unclear. Does not show a working sequential probabilistic system.
Professional Standards (5%)					
Individual Component	Exemplary standard of academic writing. Adherence to the 1,500-word limit, formatting, IEEE citation, and AI policy guidelines (Full AI declaration and appendix included). Self-Assessment included.	Professional writing. Minor errors in formatting, word count, submission, selfassessment, or citation. AI usage is correctly declared.	Satisfactory standard of writing. Understandable but with several errors in formatting, submission guidelines, or incomplete AI logs.	The report is difficult to read or understand. Significant deviations from submission requirements, word counts, or minimal AI policy adherence.	Does not meet professional standards. Ignores submission instructions, academic standards, or is missing required AI declarations.

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Project Management and Workflow (13%)					
Team Component	Excellent project management. Extremely wellstructured A2_YourFAN directory, robust handling of complexity trade-offs, complete README/requirements files, and clear evidence of proactive task planning.	Strong project management. Well-organised submission structure, logical technical decisions regarding model scalability, and clear evidence of good planning. Includes required system files.	Satisfactory project management. Meets basic structural requirements with adequate planning for the sequential inference pipeline.	Minimal project management. Disorganised file structure, poor justification of technical choices, or lack of logical workflow.	No evidence of project management. Fails to follow directory requirements and lacks structural organisation.
Team Collaboration and Integration (20%)					
Team Component	Excellent Integration Statement (under 200 words). Deep analysis of how the individual Markov module combines into the overarching high-dimensional inference engine through strict data contracts.	Clear and comprehensive Integration Statement. Strong alignment between the individual module's sequential predictions and the overarching team network architecture and data contracts.	Adequate Integration Statement. Outlines the individual role, but lacks deep analysis of how probabilities and uncertainty metrics are passed between overarching modules.	Basic integration statement. Mentions the team context but provides vague or insufficient explanation of module integration and the capstone system architecture.	Missing or inadequate integration statement. No evidence of team alignment or understanding of the broader probabilistic system.

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Self-Assessment Comments

- Individual Component – Technical Report: Well justified use case for sequential Markov modelling over i.i.d. backed by multiple research papers. Evaluated the Markov model and how its algorithms are implemented and the related computational complexities. Gave in depth analysis into results and scored against a variety of metrics.
- Individual Component – Technical Implementation: Code was modular of good quality and followed good commenting and naming conventions for ease of use and readability. Made good use of existing probabilistic ML libraries like HMMLearn.
- Individual Component – Video Demonstration: Covered everything but production quality is lacking.
- Individual Component – Professional Standards: All professional standards were met; document was formatted as per the task sheet requirements. AI declaration included, word limit adhered to, IEEE citation style.
- Team Component – Project Management and Workflow: Followed the exact structure outlined in task sheet/rubric for my directory/submodule.
- Team Component – Team Collaboration and Integration: Team integration statement clearly states how the individual Markov model integrates into the larger overarching inference engine and how strict data contracts are adhered to. Under 200 words.